Assignment

Bachelor of Computer Applications

Semester III

Database Management System

Topic: Sample Data Insertion

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Sample Data Insertion

1. Objective

To insert realistic sample data into all the tables of the Hospital Management System, ensuring that the records follow the defined primary key, foreign key, UNIQUE, CHECK, and DEFAULT constraints.

2. Introduction

Sample data is used to test whether a database works correctly. In the Hospital Management System, sample records represent departments, doctors, patients, appointments, rooms, admissions, prescriptions, medicines, laboratory tests, and bills.

The sample records were inserted using SQL INSERT INTO statements. The inserted records follow the relationships between tables. For example, appointments are associated with existing patients and doctors, while prescriptions refer to existing appointments, patients, and doctors.

3. Sample Data Insertion

Sample data was inserted into all 11 tables of the Hospital Management System.

Sl. No.	Table Name	Number of Records
1	Department	3
2	Doctor	4
3	Patient	5
4	Appointment	5
5	Room	3
6	Admission	2
7	Prescription	2
8	Medicine	3
9	Prescription_Detail	3
10	Lab_Test	3
11	Bill	3

4. Description of the Inserted Data

4.1 Department: Three departments were added: General Medicine, Cardiology, and Pediatrics. Each department has details such as its floor number and contact number.

4.2 Doctor: Four doctors were added with their names, specializations, department IDs, contact details, experience, and consultation fees.

4.3 Patient: Five patient records were added with details such as date of birth, gender, phone number, address, blood group, and emergency contact.

4.4 Appointment: Five appointments were added with patient IDs, doctor IDs, appointment dates, times, status, and reasons for consultation.

4.5 Room: Three hospital rooms were added with room numbers, room types, floor numbers, daily charges, and availability status.

4.6 Admission: Two admission records were added. They represent patients who were admitted for medical observation, including one discharged patient.

4.7 Prescription: Two prescription records were inserted with the related appointment, patient, doctor, issue date, and medical notes.

4.8 Medicine: Three medicines were added: Paracetamol 500 mg, Vitamin C Tablets, and Antacid Tablets. Their manufacturers, prices, and stock quantities were recorded.

4.9 Prescription_Detail: Three prescription-detail records were added to connect prescriptions with medicines and store dosage, duration, and quantity.

4.10 Lab_Test: Three laboratory test records were added for ECG, CBC, and Blood Glucose, including the test date, result, and status.

4.11 Bill: Three billing records were added with patient details, related appointments or admissions, bill dates, total amounts, payment statuses, and payment modes.

5. Verification of Inserted Data

The inserted data was verified using the existing SELECT * statements in the SQL file. These statements display all records from the respective tables.

department_id	department_name	floor_no	phone_no	head_doctor_id
1	General Medicine	1	08040000001	1
7001	Cardiology	2	08040007001	7001
7002	Pediatrics	3	08040007002	7002

Department table output.

patient_id	name	dob	gender	phone_no	address	blood_group	emergency_contact
1	Test Patient	2000-01-15	Male	9876500001	Bengaluru	O+	9876500002
7001	Aarav Mehta	2002-05-12	Male	9876547001	Whitefield, Bengaluru	A+	9876548001
7002	Diya Nair	2003-08-20	Female	9876547002	Marathahalli, Bengaluru	B+	9876548002
7003	Kiran Kumar	2001-11-05	Male	9876547003	Anekal, Bengaluru	O-	9876548003
7004	Meera Joseph	2004-02-18	Female	9876547004	Whitefield, Bengaluru	AB+	9876548004

Patient table output.

appointment_id	patient_id	doctor_id	appointment_date	appointment_time	status	reason
7001	7001	7001	2026-10-26	10:00:00	Scheduled	Heart checkup
7002	7002	7002	2026-10-26	11:30:00	Scheduled	Child health checkup
7003	7003	7003	2026-10-27	09:30:00	Completed	General checkup
7004	7004	7001	2026-10-28	14:00:00	Scheduled	Cardiac consultation
9002	1	1	2026-10-25	10:00:00	Scheduled	General consultation

Appointment table output.

room_id	room_number	room_type	floor_no	daily_charge	status
7001	C-201	Private	2	3000.00	Available
7002	P-301	General	3	1000.00	Available
7003	G-102	General	1	1200.00	Available

Room table output.

admission_id	patient_id	room_id	doctor_id	admission_date	discharge_date	admission_reason	status
7001	7001	7001	7001	2026-10-20	NULL	Cardiac observation	Admitted
7002	7003	7003	7003	2026-10-18	2026-10-20	Medical observation	Discharged

Admission table output.

prescription_id	appointment_id	patient_id	doctor_id	date_issued	notes
7001	7001	7001	7001	2026-10-26	Daily medication and follow-up
7002	7002	7002	7002	2026-10-26	Medication as prescribed

Prescription table output.

medicine_id	medicine_name	manufacturer	unit_price	stock_quantity
7001	Paracetamol 500 mg	ABC Pharmaceuticals	2.50	500
7002	Vitamin C Tablets	HealthCare Labs	5.00	300
7003	Antacid Tablets	MediCare Pharma	3.50	250

Medicine table output.

prescription_detail_id	prescription_id	medicine_id	dosage	duration_days	quantity
7001	7001	7002	One tablet daily	10	10
7002	7002	7001	As directed	3	6
7003	7002	7003	As directed	5	10

Prescription_Detail table output.

test_id	patient_id	doctor_id	test_name	test_date	result	status
7001	7001	7001	ECG	2026-10-26	Awaiting report	Pending
7002	7002	7002	CBC	2026-10-26	Within expected range	Completed
7003	7003	7003	Blood Glucose	2026-10-27	Awaiting report	Pending

Lab_Test table output.

bill_id	patient_id	appointment_id	admission_id	bill_date	total_amount	payment_status	payment_mode
7001	7001	7001	7001	2026-10-26	3800.00	Pending	UPI
7002	7002	7002	NULL	2026-10-26	650.00	Paid	Cash
7003	7003	7003	7002	2026-10-20	2400.00	Paid	Card

Bill table output.

Doctor table: The existing SQL file inserts four doctor records, but it does not contain a separate SELECT * FROM Doctor; statement immediately after the insertion. If you already captured the Doctor output, use that screenshot. Do not add a new query for this assignment.

The SQL file also contains an existing query that counts records in all 11 tables. Its output can be used as an additional screenshot to show the number of records in each table.

6. Data Integrity and Relationships

The sample records were designed to follow the database constraints and relationships:

1. **Primary keys:** Each record has a unique identifier in its table.
2. **Foreign keys:** Related records use IDs from existing parent records. For example, appointments refer to existing patients and doctors.
3. **UNIQUE constraints:** Department names, doctor email addresses, and room numbers must satisfy the defined uniqueness rules.
4. **CHECK constraints:** Values such as consultation fees, medicine prices, stock quantities, and status fields must follow the defined conditions.
5. **DEFAULT constraints:** Default values are defined for appointment status, room status, admission status, laboratory test status, and bill payment status.

The inserted sample records use valid values for the defined constraints.

7. Conclusion

Sample data was inserted into all 11 tables of the Hospital Management System. The records represent common hospital activities, including patient registration, doctor appointments, room admissions, prescriptions, medicine details, laboratory tests, and billing.

The existing SQL SELECT statements were used to verify the inserted records. This sample data helps demonstrate how the database stores related information and supports the operations of a hospital.